



Red Hat AI 3

Validated models

Red Hat AI validated models

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Abstract

Learn about the validated models that you can inference serve with Red Hat AI.

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PREFACE

Red Hat AI validated and enabled models have been tested and verified to work with Red Hat AI Inference. You can deploy these models for inference serving on supported hardware configurations.

CHAPTER 1. RED HAT AI VALIDATED MODELS

Red Hat AI validated models have been tested and verified to work correctly across supported hardware and product configurations. These models are available as Hugging Face downloads, as OCI artifact images, and as modelcar container images. Platform-specific validated models are also available for IBM Snyre on IBM Power and IBM Z systems.

In addition to validated models, Red Hat ships *enabled models* as modelcar container images. Enabled models are architecturally supported but have not completed the full validation pipeline. For details about the difference between validated and enabled models, see [Model support levels](#).



NOTE

If you are using AI Inference with Podman as part of a RHEL AI deployment, use ModelCar container images or Hugging Face models.

If you are using AI Inference as part of an Red Hat OpenShift AI deployment on OpenShift Container Platform, use OCI artifact images.

Red Hat uses [GuideLLM](#) for performance benchmarking and [Language Model Evaluation Harness](#) for accuracy evaluations.

For a complete list of models with platform compatibility data, see [Model support matrix](#).



IMPORTANT

AMD GPUs support only FP8 and GGUF quantization variant models. For more information, see [Supported hardware](#).

CHAPTER 2. VALIDATED MODEL SUPPORT LEVELS

Red Hat AI ships models at two support levels: validated and enabled. Understanding these support levels helps you make informed decisions about which models to deploy for your inference workloads.

Validated models

Red Hat has tested validated models with [GuideLLM](#) performance benchmarking and [Language Model Evaluation Harness](#) accuracy evaluations across specific OpenShift Container Platform, Red Hat OpenShift AI, and Red Hat AI Inference version combinations.

Validated models are benchmarked for specific use cases. This can include inference performance, quality, and other benchmarks. All third-party models are governed by the third-party license of the original model provider.

Validated models include general-purpose large language models such as Llama, Granite, Mistral, Qwen, and Phi model families, and quantized variants in FP8, INT4, INT8, NVFP4, and BF16 formats.

Enabled models

Red Hat ships enabled models as modelcar container images with architecturally compatible configurations. Enabled models have not completed the full benchmarking and accuracy evaluation pipeline that validated models receive.

Enabled models include specialty categories such as:

- Embedding models, for example **granite-embedding-english-r2**, **all-MiniLM-L6-v2**, **nomic-embed-text-v1.5**, and **Qwen3-Embedding-8B**
- Safety and guard models, for example **Llama-Guard-4-12B** and **granite-guardian-3.2-5b**
- Security models, for example **Foundation-Sec-8B-Instruct**
- Reasoning models, for example **Phi-4-reasoning**
- Additional general-purpose models not yet through the full validation pipeline

Both support levels indicate that Red Hat ships the model and provides support. The key difference is the depth of testing: validated models have quantified performance and accuracy data for specific platform configurations, while Red Hat verifies that enabled models work with the inference server architecture.

To find the support level for a specific model, see [Model support matrix](#).

CHAPTER 3. VALIDATED MODEL SUPPORT MATRIX

You can use the model support matrix to verify that a model is compatible with your Red Hat AI Inference, Red Hat OpenShift AI, and vLLM version combination before deploying it for inference serving. The matrix lists all validated and enabled models with their minimum platform version requirements and modelcar container image paths.

For more information, see [Red Hat AI models](#) on Hugging Face.



NOTE

Verify that your deployed Red Hat AI Inference, Red Hat OpenShift AI, and vLLM versions meet or exceed the minimum versions listed for your target model. For an explanation of the Validated and Enabled status values, see [Model support levels](#).



NOTE

Hugging Face links require internet access. If you are working in a disconnected environment, use the modelcar container image paths with your mirrored registry. For more information, see [Deploying the standalone Red Hat AI Inference container in a disconnected environment](#).

Table 3.1. Red Hat AI Model support matrix

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/granite-3.1-8b-instruct	registry.redhat.io/rhelai/modelcar-granite-3-1-8b-instruct:1.5	Validated	v0.8.4	3	2.21	19 GB	1XA100-40, 1XA100-80, 1XH100, 1XH200, 1XL4, 2XA100-40, 2XA100-80, 2XH100, 2XL4, 4XA100-40, 4XA100-80, 4XH100, 4XL4, 8XA100-40, 8XA100-80	n/a
RedHatAI/granite-4.0-h-tiny-FP8-dynamic	registry.redhat.io/rhai/modelcar-granite-4-0-h-tiny-fp8-dynamic:3.0	Validated	v0.13.0	3.3.0	3.3.0	9 GB	1XB200, 1XH100, 1XH200, 1XL4	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/Llama-3.1-8B-Instruct	registry.redhat.io/rhelaii/modelcar-llama-3-1-8b-instruct:1.5	Validated	v0.8.4	3	2.21	19 GB	1XA100-40, 1XA100-80, 1XH100, 1XH200, 1XL4, 2XA100-40, 2XA100-80, 2XH100, 2XL4, 4XA100-40, 4XA100-80, 4XH100, 4XL4, 8XA100-40, 8XA100-80	n/a
RedHatAI/Llama-3.3-70B-Instruct	registry.redhat.io/rhelaii/modelcar-llama-3-3-70b-instruct:1.5	Validated	v0.8.4	3	2.21	163 GB	2XH200, 4XA100-80, 4XH100, 8XA100-40, 8XH100	n/a
RedHatAI/Llama-4-Maverick-17B-128E-Instruct	registry.redhat.io/rhelaii/modelcar-llama-4-maverick-17b-128e-instruct:1.5	Validated	v0.8.4	3	2.21	924 GB	8XH200	n/a
RedHatAI/Llama-4-Maverick-17B-128E-Instruct-FP8	registry.redhat.io/rhelaii/modelcar-llama-4-maverick-17b-128e-instruct-fp8:1.5	Validated	v0.8.4	3	2.21	480 GB	4XH200, 8XH100	n/a
RedHatAI/Llama-4-Scout-17B-16E-Instruct	registry.redhat.io/rhelaii/modelcar-llama-4-scout-17b-16e-instruct:1.5	Validated	v0.8.4	3	2.21	250 GB	4XA100-80, 4XH100, 4XH200, 8XH100	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/phi-4	registry.redhat.io/rhelai/modelcar-phi-4:1.5	Validated	v0.8.4	3	2.21	34 GB	1XA100-80, 1XH100, 1XH200, 2XA100-40, 2XA100-80, 2XH100, 2XL4	n/a
RedHatAI/Devstral-Small-2-24B-Instruct-2512	registry.redhat.io/rhai/modelcar-devstral-small-2-24b-instruct-2512:3.0	Validated	v0.14.1	3.4.0- ea.1	3.4.0- ea.1	30 GB	1XA100-80, 1XB200, 1XH100, 1XH200, 2XA100-80, 2XB200, 2XH100, 2XH200, 4XA100-80, 4XB200, 4XH100, 4XH200, 4XL4, 8XA100-80, 8XB200, 8XH100, 8XH200, 8XL4	n/a
RedHatAI/Ministral-3-3B-Instruct-2512	registry.redhat.io/rhai/modelcar-ministral-3-3b-instruct-2512:3.0	Validated	v0.14.1	3.4.0- ea.1	3.4.0- ea.1	6 GB	1XA100-80, 1XB200, 1XH100, 1XH200, 1XL4, 2XA100-80, 2XB200, 2XH100, 2XH200, 2XL4, 4XA100-80, 4XB200, 4XH100, 4XH200, 4XL4, 8XA100-80, 8XB200, 8XH100, 8XH200, 8XL4	n/a
RedHatAI/Mistral-Large-3-675B-Instruct-2512	registry.redhat.io/rhai/modelcar-mistral-large-3-675b-instruct-2512:3.0	Validated	v0.11.2	3.2.5	3.2	784 GB	8XB200, 8XH200	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/Mistral-Large-3-675B-Instruct-2512-NVFP4	registry.redhat.io/rhai/modelcar-mistral-large-3-675b-instruct-2512-nvfp4:3.0	Validated	v0.11.2	3.2.5	3.2	464 GB	4XB200, 4XH200, 8XA100-80, 8XB200, 8XH100, 8XH200	n/a
RedHatAI/Mistral-Small-24B-Instruct-2501	registry.redhat.io/rhelaii/modelcar-mistral-small-24b-instruct-2501:1.5	Validated	v0.8.4	3	2.21	55 GB	1XA100-80, 1XH100, 2XA100-40, 2XA100-80, 2XH100, 4XA100-40, 4XA100-80, 4XH100, 4XL4, 8XA100-40, 8XH100	n/a
RedHatAI/Mistral-Small-3.1-24B-Instruct-2503	registry.redhat.io/rhelaii/modelcar-mistral-small-3-1-24b-instruct-2503:1.5	Validated	v0.8.4	3	2.21	56 GB	1XA100-80, 1XH100, 1XH200, 2XA100-40, 2XA100-80, 2XH100, 4XA100-40, 4XA100-80, 4XH100, 4XL4, 8XA100-40, 8XH100	n/a
RedHatAI/Mixtral-8x7B-Instruct-v0.1	registry.redhat.io/rhelaii/modelcar-mixtral-8x7b-instruct-v0-1:1.4	Validated	v0.8.4	3	2.21	108 GB	1XH200, 2XA100-80, 2XH100, 4XA100-40, 4XA100-80, 4XH100, 8XA100-40, 8XH100, 8XL4	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/NVIDIA-Nemotron-Nano-9B-v2-quantized.w4a16	registry.redhat.io/rhelai/modelcar-nvidia-nemotron-nano-9b-v2-quantized-w4a16:1.5	Validated	v0.11.0	3.2.3, 3.2.4	3	8 GB	1XH100	n/a
RedHatAI/Llama-3.1-Nemotron-70B-Instruct-HF	registry.redhat.io/rhelai/modelcar-llama-3-1-nemotron-70b-instruct-hf:1.5	Validated	v0.8.4	3	2.21	163 GB	2XH200, 4XA100-80, 4XH100, 8XA100-40	n/a
RedHatAI/NVIDIA-Nemotron-3-Nano-30B-A3B-FP8	registry.redhat.io/rhai/modelcar-nvidia-nemotron-3-nano-30b-a3b-fp8:3.0	Validated	v0.11.2	3.2.5	3.2	38 GB	1XB200, 1XH100, 1XH200, 2XB200, 2XH100, 2XH200, 4XB200, 4XH100, 4XH200, 8XB200, 8XH100, 8XH200	n/a
RedHatAI/NVIDIA-Nemotron-3-Super-120B-A12B-BF16	registry.redhat.io/rhai/modelcar-nvidia-nemotron-3-super-120b-a12b-bf16:3.0	Validated	v0.17.1	3.4.0- ea.2	3.4.0- ea.2	285 GB	2XH200, 4XA100-80, 4XH100, 4XH200, 8XA100-80, 8XH100, 8XH200	n/a
RedHatAI/NVIDIA-Nemotron-3-Super-120B-A12B-FP8	registry.redhat.io/rhai/modelcar-nvidia-nemotron-3-super-120b-a12b-fp8:3.0	Validated	v0.17.1	3.4.0- ea.2	3.4.0- ea.2	148 GB	1XH200, 2XH100, 2XH200, 4XH100, 4XH200, 8XH100, 8XH200	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/NVIDIA-Nemotron-3-Super-120B-A12B-NVFP4	registry.redhat.io/rhai/modelcar-nvidia-nemotron-3-super-120b-a12b-nvfp4:3.0	Validated	v0.17.1	3.4.0- ea.2	3.4.0- ea.2	93 GB	1XH200, 2XH100, 2XH200	n/a
RedHatAI/gpt-oss-120b	registry.redhat.io/rhelai/modelcar-gpt-oss-120b:1.5	Validated	v0.10.1	3.2.2	2.25	76 GB	1XB200, 1XH100, 1XH200, 2XB200, 2XH100, 2XH200, 4XA100-40, 4XB200, 4XH100, 4XH200, 4XL4, 8XA100-40, 8XB200, 8XH100, 8XH200, 8XL4	n/a
RedHatAI/gpt-oss-20b	registry.redhat.io/rhelai/modelcar-gpt-oss-20b:1.5	Validated	v0.10.1	3.2.2	2.25	16 GB	1XA100-40, 1XB200, 1XH100, 1XH200, 1XL4, 2XA100-40, 2XB200, 2XH100, 2XH200, 2XL4, 4XA100-40, 4XB200, 4XH100, 4XH200, 4XL4, 8XA100-40, 8XB200, 8XH100, 8XH200, 8XL4	n/a
RedHatAI/Qwen2.5-7B-Instruct	registry.redhat.io/rhelai/modelcar-qwen2-5-7b-instruct:1.5	Validated	v0.8.4	3	2.21	18 GB	1XA100-40, 1XA100-80, 1XH100, 1XH200, 1XL4, 2XA100-40, 2XA100-80, 2XH100, 2XL4, 4XA100-40, 4XA100-80, 4XH100	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
Qwen/Qwen3-8B-FP8	registry.redhat.io/rhelai/modelcar-qwen3-8b-fp8:1.5	Validated	v0.10.0	3.2.1	2.24	11 GB	1XA100-40, 1XH100, 1XL4, 2XH100	n/a
RedHatAI/Apertus-8B-Instruct-2509-FP8-dynamic	registry.redhat.io/rhai/modelcar-apertus-8b-instruct-2509-fp8-dynamic:3.0	Validated	v0.11.2	3.2.5	3.2	11 GB	1XA100-80, 1XB200, 1XH100, 1XH200, 2XA100-80, 2XB200, 2XH100, 2XH200, 4XA100-80, 4XB200, 4XH100, 4XH200, 8XA100-80, 8XB200, 8XH100, 8XH200	n/a
RedHatAI/DeepSeek-R1-0528-quantized.w4a16	registry.redhat.io/rhelai/modelcar-deepseek-r1-0528-quantized-w4a16:1.5	Validated	v0.10.0	3.2.1	2.24	428 GB	4XB200, 4XH200, 8XB200, 8XH100, 8XH200	n/a
RedHatAI/gemma-3n-E4B-it-FP8-dynamic	registry.redhat.io/rhelai/modelcar-gemma-3n-e4b-it-fp8-dynamic:1.5	Validated	v0.10.0	3.2.1	2.24	14 GB	1XA100-40, 1XH100, 1XH200, 1XL4, 2XA100-40, 2XH100, 2XH200, 2XL4, 4XA100-40, 4XH100, 4XH200, 4XL4, 8XA100-40, 8XH100, 8XH200, 8XL4	n/a
RedHatAI/granite-3.1-8b-instruct-fp8-dynamic	registry.redhat.io/rhelai/modelcar-granite-3-1-8b-instruct-fp8-dynamic:1.5	Validated	v0.8.4	3	2.21	11 GB	1XH200	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RHAI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/granite-3.1-8b-instruct-quantized.w4a16	registry.redhat.io/rhelai/modelcar-granite-3-1-8b-instruct-quantized-w4a16:1.5	Validated	v0.8.4	3	2.21	6 GB	1XH200	n/a
RedHatAI/granite-4.0-h-small-FP8-dynamic	registry.redhat.io/rhai/modelcar-granite-4-0-h-small-fp8-dynamic:3.0	Validated	v0.13.0	3.3.0	3.3.0	38 GB	1XA100-80, 1XB200, 1XH100, 1XH200	n/a
RedHatAI/Kimi-K2-Instruct-quantized.w4a16	registry.redhat.io/rhelai/modelcar-kimi-k2-instruct-quantized-w4a16:1.5	Validated	v0.10.0	3.2.1	2.24	629 GB	4XB200, 8XB200, 8XH200	n/a
RedHatAI/Llama-3.1-Nemotron-70B-Instruct-HF-FP8-dynamic	registry.redhat.io/rhelai/modelcar-llama-3-1-nemotron-70b-instruct-hf-fp8-dynamic:1.5	Validated	v0.8.4	3	2.21	84 GB	1XH200, 2XA100-80, 2XH100, 4XA100-40, 4XA100-80, 4XH100, 8XA100-40, 8XH100	n/a
RedHatAI/Llama-3.3-70B-Instruct-FP8-dynamic	registry.redhat.io/rhelai/modelcar-llama-3-3-70b-instruct-fp8-dynamic:1.5	Validated	v0.8.4	3	2.21	84 GB	1XH200, 2XH100, 4XA100-40, 4XA100-80, 4XH100, 8XA100-40, 8XA100-80, 8XH100	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/Llama-3.3-70B-Instruct-quantized.w4a16	registry.redhat.io/rhelai1/modelcar-llama-3-3-70b-instruct-quantized-w4a16:1.5	Validated	v0.8.4	3	2.21	46 GB	1XH100, 2XH100, 4XA100-40, 4XH100, 8XA100-40, 8XH100	n/a
RedHatAI/Llama-3.3-70B-Instruct-quantized.w8a8	registry.redhat.io/rhelai1/modelcar-llama-3-3-70b-instruct-quantized-w8a8:1.5	Validated	v0.8.4	3	2.21	84 GB	2XH100, 4XA100-40, 4XA100-80, 4XH100, 8XA100-40, 8XH100	n/a
RedHatAI/Llama-4-Scout-17B-16E-Instruct-FP8-dynamic	registry.redhat.io/rhelai1/modelcar-llama-4-scout-17b-16e-instruct-fp8-dynamic:1.5	Validated	v0.8.4	3	2.21	132 GB	2XH100, 2XH200, 4XH100, 8XH100, 8XL4	n/a
RedHatAI/Llama-4-Scout-17B-16E-Instruct-quantized.w4a16	registry.redhat.io/rhelai1/modelcar-llama-4-scout-17b-16e-instruct-quantized-w4a16:1.5	Validated	v0.8.4	3	2.21	75 GB	2XH100, 2XH200, 4XA100-40, 4XH100, 8XA100-40, 8XH100	n/a
RedHatAI/Meta-Llama-3.1-8B-Instruct-FP8-dynamic	registry.redhat.io/rhelai1/modelcar-llama-3-1-8b-instruct-fp8-dynamic:1.5	Validated	v0.8.4	3	2.21	11 GB	1XH200	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/MiniMax-M2.5	registry.redhat.io/rhai/modelcar-minimax-m2-5:3.0	Validated	v0.14.1	3.4.0- ea.1	3.4.0- ea.1	265 GB	2XB200, 4XA100-80, 4XB200, 4XH100, 4XH200	n/a
RedHatAI/Mistral-3-14B-Instruct-2512	registry.redhat.io/rhai/modelcar-mistral-3-14b-instruct-2512:3.0	Validated	v0.13.0	3.3.0	3.3.0	19 GB	1XA100-80, 1XB200, 1XH100, 1XH200, 1XL4, 2XA100-80, 2XB200, 2XH100, 2XH200, 2XL4, 4XA100-80, 4XB200, 4XH100, 4XH200, 4XL4, 8XA100-80, 8XB200, 8XH100, 8XH200, 8XL4	n/a
RedHatAI/Mistral-Small-3.1-24B-Instruct-2503-FP8-dynamic	registry.redhat.io/rhelaii/modelcar-mistral-small-3-1-24b-instruct-2503-fp8-dynamic:1.5	Validated	v0.8.4	3	2.21	30 GB	1XA100-80, 1XH100, 1XH200, 2XA100-40, 2XA100-80, 2XH100, 2XL4, 4XA100-40, 4XA100-80, 4XH100, 4XL4, 8XA100-40, 8XH100	n/a
RedHatAI/Mistral-Small-3.1-24B-Instruct-2503-quantized.w4a16	registry.redhat.io/rhelaii/modelcar-mistral-small-3-1-24b-instruct-2503-quantized-w4a16:1.5	Validated	v0.8.4	3	2.21	18 GB	1XA100-40, 1XA100-80, 1XH100, 1XH200, 2XA100-40, 2XA100-80, 2XH100, 4XA100-40, 4XA100-80, 4XH100, 8XA100-40, 8XH100	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/Mistral-Small-3.1-24B-Instruct-2503-quantized.w8a8	registry.redhat.io/rhelai/modelcar-mistral-small-3-1-24b-instruct-2503-quantized-w8a8:1.5	Validated	v0.8.4	3	2.21	30 GB	1XA100-40, 1XA100-80, 1XH100, 1XH200, 2XA100-40, 2XA100-80, 2XH100, 2XL4, 4XA100-40, 4XA100-80, 4XH100, 4XL4, 8XA100-40, 8XA100-80, 8XH100	n/a
RedHatAI/NVIDIA-Nemotron-Nano-9B-v2-FP8-dynamic	registry.redhat.io/rhelai/modelcar-nvidia-nemotron-nano-9b-v2-fp8-dynamic:1.5	Validated	v0.10.1	3.2.2	2.25	12 GB	1XA100-40, 1XB200, 1XH100, 1XH200	n/a
RedHatAI/phi-4-FP8-dynamic	registry.redhat.io/rhelai/modelcar-phi-4-fp8-dynamic:1.5	Validated	v0.8.4	3	2.21	19 GB	1XA100-40, 1XA100-80, 1XH100, 1XH200, 2XA100-40, 2XA100-80, 2XH100, 2XL4	n/a
RedHatAI/Phi-4-mini-instruct-FP8-dynamic	registry.redhat.io/rhelai/modelcar-phi-4-mini-instruct-fp8-dynamic:1.5	Validated	v0.14.1	3.4.0- ea.1	3.4.0- ea.1	7 GB	1XA100-80, 1XB200, 1XH100, 1XH200, 1XL4, 2XA100-80, 2XB200, 2XH100, 2XH200, 2XL4, 4XA100-80, 4XB200, 4XH100, 4XH200, 4XL4, 8XA100-80, 8XH100, 8XH200, 8XL4	n/a
RedHatAI/phi-4-quantized.w4a16	registry.redhat.io/rhelai/modelcar-phi-4-quantized-w4a16:1.5	Validated	v0.8.4	3	2.21	11 GB	1XA100-40, 1XA100-80, 1XH100, 2XA100-40, 2XA100-80, 2XH100, 2XL4	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RHAI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/phi-4-quantized.w8a8	registry.redhat.io/rhelai/modelcar-phi-4-quantized-w8a8:1.5	Validated	v0.8.4	3	2.21	19 GB	1XA100-40, 1XA100-80, 1XH100, 2XA100-40, 2XA100-80, 2XH100	n/a
RedHatAI/Phi-4-reasoning-FP8-dynamic	registry.redhat.io/rhai/modelcar-phi-4-reasoning-fp8-dynamic:3.0	Validated	v0.13.0	3.3.0	3.3.0	19 GB	1XA100-80, 1XB200, 1XH100, 1XH200, 1XL4, 2XA100-40, 2XA100-80, 2XB200, 2XH100, 2XH200, 2XL4	n/a
RedHatAI/Qwen2.5-7B-Instruct-FP8-dynamic	registry.redhat.io/rhelai/modelcar-qwen2-5-7b-instruct-fp8-dynamic:1.5	Validated	v0.8.4	3	2.21	11 GB	1XA100-40, 1XA100-80, 1XH100, 1XH200, 1XL4, 2XA100-40, 2XA100-80, 2XH100, 2XL4, 4XA100-40, 4XA100-80, 4XH100	n/a
RedHatAI/Qwen2.5-7B-Instruct-quantized.w4a16	registry.redhat.io/rhelai/modelcar-qwen2-5-7b-instruct-quantized-w4a16:1.5	Validated	v0.8.4	3	2.21	7 GB	1XA100-40, 1XA100-80, 1XH100, 1XL4, 2XA100-40, 2XH100, 4XA100-40, 4XH100, 4XL4	n/a
RedHatAI/Qwen2.5-7B-Instruct-quantized.w8a8	registry.redhat.io/rhelai/modelcar-qwen2-5-7b-instruct-quantized-w8a8:1.5	Validated	v0.8.4	3	2.21	11 GB	1XA100-40, 1XA100-80, 1XH100, 1XL4, 2XA100-40, 2XA100-80, 2XH100, 2XL4, 4XA100-40, 4XH100, 4XL4	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/Qwen3.5-122B-A10B-FP8-dynamic	registry.redhat.io/rhai/modelcar-qwen3-5-122b-a10b-fp8-dynamic:3.0	Validated	v0.17.1	3.4.0- ea.2	3.4.0- ea.2	148 GB	2XA100-80, 2XH100, 4XA100-80, 4XH200, 8XA100-80, 8XH100	n/a
RedHatAI/Qwen3.5-35B-A3B-FP8-dynamic	registry.redhat.io/rhai/modelcar-qwen3-5-35b-a3b-fp8-dynamic:3.0	Validated	v0.17.1	3.4.0- ea.2	3.4.0- ea.2	44 GB	1XA100-80, 1XH100, 1XH200, 2XA100-80, 2XH100, 2XH200, 4XA100-80, 4XH100, 4XH200, 8XA100-80, 8XH100, 8XH200	n/a
RedHatAI/Qwen3.5-397B-A17B-FP8-dynamic	registry.redhat.io/rhai/modelcar-qwen3-5-397b-a17b-fp8-dynamic:3.0	Validated	v0.17.1	3.4.0- ea.2	3.4.0- ea.2	466 GB	4XH200, 8XA100-80, 8XH100	n/a
RedHatAI/Qwen3-8B-FP8-dynamic	registry.redhat.io/rhelai/modelcar-qwen3-8b-fp8-dynamic:1.5	Validated	v0.10.0	3.2.1	2.24	11 GB	1XA100-40, 1XH100, 1XH200, 1XL4, 2XA100-40, 2XH100, 2XH200, 2XL4, 4XA100-40, 4XH100, 4XH200, 4XL4, 8XA100-40, 8XH100, 8XH200, 8XL4	n/a
RedHatAI/Qwen3-Coder-480B-A35B-Instruct-FP8	registry.redhat.io/rhelai/modelcar-qwen3-coder-480b-a35b-instruct-fp8:1.5	Validated	v0.10.1	3.2.2	2.25	555 GB	4XB200, 4XH200	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RHAI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/Qwen3-Coder-Next-NVFP4	registry.redhat.io/rhai/modelcar-qwen3-coder-next-nvfp4:3.0	Validated	v0.14.1	3.4.0- ea.1	3.4.0- ea.1	55 GB	1XB200, 1XH100, 1XH200, 2XB200, 2XH100, 2XH200, 4XB200, 4XH100, 4XH200, 8XB200, 8XH100, 8XH200	n/a
RedHatAI/Qwen3-Next-80B-A3B-Instruct-FP8	registry.redhat.io/rhai/modelcar-qwen3-next-80b-a3b-instruct-fp8:3.0	Validated	v0.13.0	3.3.0	3.3.0	95 GB	1XB200, 2XA100-80, 2XB200, 2XH200, 4XA100-80, 4XH200	n/a
RedHatAI/Qwen3-Next-80B-A3B-Instruct-quantized.w4a16	registry.redhat.io/rhai/modelcar-qwen3-next-80b-a3b-instruct-quantized-w4a16:3.0	Validated	v0.13.0	3.3.0	3.3.0	51 GB	1XA100-80, 1XB200, 1XH100, 1XH200, 2XA100-80, 2XB200, 2XH100, 2XH200, 4XA100-80, 4XB200, 4XH100, 4XH200	n/a
RedHatAI/Qwen3-VL-235B-A22B-Instruct-NVFP4	registry.redhat.io/rhai/modelcar-qwen3-vl-235b-a22b-instruct-nvfp4:3.0	Validated	v0.13.0	3.3.0	3.3.0	156 GB	1XB200, 2XA100-80, 2XB200, 2XH100, 2XH200, 4XA100-80, 4XB200, 4XH100, 4XH200, 8XA100-80, 8XB200, 8XH100, 8XH200	n/a
RedHatAI/sarvam-105b-FP8-Dynamic	registry.redhat.io/rhai/modelcar-sarvam-105b-fp8-dynamic:3.0	Validated	v0.18.0	3.4.0	3.4.0	130 GB	1XH200, 2XH200, 4XH100, 4XH200, 8XH100, 8XH200	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/sarvam-30b-FP8-Dynamic	registry.redhat.io/rhai/modelcar-sarvam-30b-fp8-dynamic:3.0	Validated	v0.18.0	3.4.0	3.4.0	45 GB	1XA100-80, 1XH100, 1XH200, 2XA100-80, 2XH100, 2XH200, 4XA100-80, 4XH100, 4XH200, 8XA100-80, 8XH100, 8XH200	n/a
RedHatAI/SmolLM3-3B-FP8-dynamic	registry.redhat.io/rhelai/modelcar-smolLM3-3b-fp8-dynamic:1.5	Validated	v0.10.11	3.2.2	2.25	5 GB	1XA100-40, 1XB200, 1XH100, 1XH200, 1XL4, 2XA100-40, 2XB200, 2XH100, 2XH200, 2XL4, 4XA100-40, 4XB200, 4XH100, 4XH200, 4XL4	n/a
RedHatAI/gpt-oss-120b-essential	registry.redhat.io/rhai/modelcar-gpt-oss-120b-essential:3.0	Validated	v0.10.11	3.2.2	2.25	76 GB	1XB200, 1XH100, 1XH200, 2XB200, 2XH100, 2XH200, 4XA100-40, 4XB200, 4XH100, 4XH200, 4XL4, 8XA100-40, 8XB200, 8XH100, 8XH200, 8XL4	n/a
RedHatAI/gpt-oss-20b-essential	registry.redhat.io/rhai/modelcar-gpt-oss-20b-essential:3.0	Validated	v0.10.11	3.2.2	2.25	16 GB	1XA100-40, 1XB200, 1XH100, 1XH200, 1XL4, 2XA100-40, 2XB200, 2XH100, 2XH200, 2XL4, 4XA100-40, 4XB200, 4XH100, 4XH200, 4XL4, 8XA100-40, 8XB200, 8XH100, 8XH200, 8XL4	n/a
google/gemma-3-1b-it	registry.redhat.io/rhai/modelcar-gemma-3-1b-it:3.0	Enabled	v0.11.2	3.2.5	3.2	3 GB	1XH200	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
google/gemma-3-27b-it	registry.redhat.io/rhai/modelcar-gemma-3-27b-it:3.0	Enabled	v0.11.2	3.2.5	3.2	64 GB	1XH200	n/a
ibm-granite/granite-guardian-3.2-5b	registry.redhat.io/rhai/modelcar-granite-guardian-3-2-5b:3.0	Enabled	v0.11.2	3.2.5	3.2	14 GB	1XH200	n/a
meta-llama/Llama-2-7b-chat-hf	registry.redhat.io/rhai/modelcar-llama-2-7b-chat-hf:3.0	Enabled	v0.11.2	3.2.5	3.2	16 GB	1XH200	n/a
RedHatAI/Foundation-Sec-8B-Instruct	registry.redhat.io/rhai/modelcar-foundation-sec-8b-instruct:3.0	Enabled	v0.13.0	3.3.0	3.3.0	19 GB	1XH200	n/a
RedHatAI/gemma-3-12b-it	registry.redhat.io/rhai/modelcar-gemma-3-12b-it:3.0	Enabled	v0.11.2	3.2.5	3.2	29 GB	1XH200	n/a
RedHatAI/gemma-4-26B-A4B-it-FP8-Dynamic	registry.redhat.io/rhai/modelcar-gemma-4-26b-a4b-it-fp8-dynamic:3.0	Enabled	v0.18.0	3.4.0	3.4.0	33 GB	1XH200	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/gemma-4-31B-it-FP8-Dynamic	registry.redhat.io/rhai/modelcar-gemma-4-31b-it-fp8-dynamic:3.0	Enabled	v0.18.0	3.4.0	3.4.0	39 GB	1XH200	n/a
RedHatAI/Llama-Guard-4-12B	registry.redhat.io/rhai/modelcar-llama-guard-4-12b:3.0	Enabled	v0.11.2	3.2.5	3.2	28 GB	1XH200	n/a
RedHatAI/Phi-4-reasoning	registry.redhat.io/rhai/modelcar-phi-4-reasoning:3.0	Enabled	v0.11.2	3.2.5	3.2	34 GB	1XH200	n/a
RedHatAI/Qwen3-Embedding-8B	registry.redhat.io/rhelai/modelcar-qwen3-embedding-8b:1.5	Enabled	v0.11.2	3.2.5	3.2	17 GB	1XL4	n/a
RedHatAI/granite-embedding-english-r2	registry.redhat.io/rhai/modelcar-granite-embedding-english-r2:3.0	Enabled	v0.11.2	3.2.5	3.2	1 GB	1XL4	n/a
RedHatAI/all-MiniLM-L6-v2	registry.redhat.io/rhai/modelcar-all-minilm-l6-v2:3.0	Enabled	v0.11.2	3.2.5	3.2	1 GB	1XL4	n/a

Model	Modelcar	Status	Min. vLLM version	Min. RH AI version	Min. RH OAI version	Min. vRAM (GB)	Supported GPUs	Migration guidance
RedHatAI/nomic-embed-text-v1.5	registry.redhat.io/rhelai1/modelcar-nomic-embed-text-v1-5:1.5	Enabled	v0.11.2	3.2.5	3.2	1 GB	1XL4	n/a

Additional resources

- [Model support levels](#)
- [Validated Red Hat AI ModelCar container images](#)
- [Validated OCI artifact model container images for OpenShift Container Platform and Red Hat OpenShift AI deployments](#)

CHAPTER 4. VALIDATED OCI ARTIFACT MODEL CONTAINER IMAGES

The following table lists validated OCI artifact model container images available from the Red Hat container registry, including baseline and quantized variants for each supported model.

Table 4.1. Validated OCI artifact model container images

Model	Quantized variants	OCI artifact images
llama-4-scout-17b-16e-instruct	INT4, FP8	- Baseline: registry.redhat.io/rhelai1/llama-4-scout-17b-16e-instruct:1.5 - INT4: registry.redhat.io/rhelai1/llama-4-scout-17b-16e-instruct-quantized-w4a16:1.5 - FP8: registry.redhat.io/rhelai1/llama-4-scout-17b-16e-instruct-fp8-dynamic:1.5
llama-4-maverick-17b-128e-instruct	FP8	- Baseline: registry.redhat.io/rhelai1/llama-4-maverick-17b-128e-instruct:1.5 - FP8: registry.redhat.io/rhelai1/llama-4-maverick-17b-128e-instruct-fp8:1.5
mistral-small-3-1-24b-instruct-2503	INT4, INT8, FP8	- Baseline: registry.redhat.io/rhelai1/mistral-small-3-1-24b-instruct-2503:1.5 - INT4: registry.redhat.io/rhelai1/mistral-small-3-1-24b-instruct-2503-quantized-w4a16:1.5 - INT8: registry.redhat.io/rhelai1/mistral-small-3-1-24b-instruct-2503-quantized-w8a8:1.5 - FP8: registry.redhat.io/rhelai1/mistral-small-3-1-24b-instruct-2503-fp8-dynamic:1.5

Model	Quantized variants	OCI artifact images
llama-3-3-70b-instruct	INT4, INT8, FP8	● Baseline: registry.redhat.io/rhelai1/llama-3-3-70b-instruct:1.5 ● INT4: registry.redhat.io/rhelai1/llama-3-3-70b-instruct-quantized-w4a16:1.5 ● INT8: registry.redhat.io/rhelai1/llama-3-3-70b-instruct-quantized-w8a8:1.5 ● FP8: registry.redhat.io/rhelai1/llama-3-3-70b-instruct-fp8-dynamic:1.5
llama-3-1-8b-instruct	INT4, INT8, FP8	● Baseline: registry.redhat.io/rhelai1/llama-3-1-8b-instruct:1.5 ● INT4: registry.redhat.io/rhelai1/llama-3-1-8b-instruct-quantized-w4a16:1.5 ● INT8: registry.redhat.io/rhelai1/llama-3-1-8b-instruct-quantized-w8a8:1.5 ● FP8: registry.redhat.io/rhelai1/llama-3-1-8b-instruct-fp8-dynamic:1.5

Model	Quantized variants	OCI artifact images
granite-3-1-8b-instruct	INT4, INT8, FP8	- Baseline: registry.redhat.io/rhelai1/granite-3-1-8b-instruct:1.5 - INT4: registry.redhat.io/rhelai1/granite-3-1-8b-instruct-quantized-w4a16:1.5 - INT8: registry.redhat.io/rhelai1/granite-3-1-8b-instruct-quantized-w8a8:1.5 - FP8: registry.redhat.io/rhelai1/granite-3-1-8b-instruct-fp8-dynamic:1.5
phi-4	INT4, INT8, FP8	- Baseline: registry.redhat.io/rhelai1/phi-4:1.5 - INT4: registry.redhat.io/rhelai1/phi-4-quantized-w4a16:1.5 - INT8: registry.redhat.io/rhelai1/phi-4-quantized-w8a8:1.5 - FP8: registry.redhat.io/rhelai1/phi-4-fp8-dynamic:1.5

Model	Quantized variants	OCI artifact images
qwen2-5-7b-instruct	INT4, INT8, FP8	- Baseline: registry.redhat.io/rhelai1/qwen2-5-7b-instruct:1.5 - INT4: registry.redhat.io/rhelai1/qwen2-5-7b-instruct-quantized-w4a16:1.5 - INT8: registry.redhat.io/rhelai1/qwen2-5-7b-instruct-quantized-w8a8:1.5 - FP8: registry.redhat.io/rhelai1/qwen2-5-7b-instruct-fp8-dynamic:1.5
mistral-small-24b-instruct-2501	INT4, INT8, FP8	- Baseline: registry.redhat.io/rhelai1/mistral-small-24b-instruct-2501:1.5 - INT4: registry.redhat.io/rhelai1/mistral-small-24b-instruct-2501-quantized-w4a16:1.5 - INT8: registry.redhat.io/rhelai1/mistral-small-24b-instruct-2501-quantized-w8a8:1.5 - FP8: registry.redhat.io/rhelai1/mistral-small-24b-instruct-2501-fp8-dynamic:1.5
mixtral-8x7b-instruct-v0-1	None	Baseline: registry.redhat.io/rhelai1/mixtral-8x7b-instruct-v0-1:1.4
granite-3-1-8b-base	INT4 (baseline currently unavailable)	INT4: registry.redhat.io/rhelai1/granite-3-1-8b-base-quantized-w4a16:1.5

Model	Quantized variants	OCI artifact images
granite-3.1-8b-starter-v2	None	Baseline: registry.redhat.io/rhelai1/granite-3.1-8b-starter-v2:1.5
llama-3-1-nemotron-70b-instruct-hf	FP8	- Baseline: registry.redhat.io/rhelai1/llama-3-1-nemotron-70b-instruct-hf:1.5 - FP8: registry.redhat.io/rhelai1/llama-3-1-nemotron-70b-instruct-hf-fp8-dynamic:1.5
gemma-2-9b-it	FP8	- Baseline: registry.redhat.io/rhelai1/gemma-2-9b-it:1.5 - FP8: registry.redhat.io/rhelai1/gemma-2-9b-it-fp8:1.5
deepseek-r1-0528	INT4 (baseline currently unavailable)	INT4: registry.redhat.io/rhelai1/deepseek-r1-0528-quantized-w4a16:1.5
qwen3-8b	FP8 (baseline currently unavailable)	FP8: registry.redhat.io/rhelai1/qwen3-8b-fp8-dynamic:1.5
kimi-k2-instruct	INT4 (baseline currently unavailable)	INT4: registry.redhat.io/rhelai1/kimi-k2-instruct-quantized-w4a16:1.5
gemma-3n-e4b-it	FP8 (baseline currently unavailable)	FP8: registry.redhat.io/rhelai1/gemma-3n-e4b-it-fp8-dynamic:1.5

Model	Quantized variants	OCI artifact images
gpt-oss-120b	None	Baseline: registry.redhat.io/rhelai1/gpt-oss-120b:1.5
gpt-oss-20b	None	Baseline: registry.redhat.io/rhelai1/gpt-oss-20b:1.5
qwen3-coder-480b-a35b-instruct	FP8 (baseline currently unavailable)	FP8: registry.redhat.io/rhelai1/qwen3-coder-480b-a35b-instruct-fp8:1.5
whisper-large-v3-turbo	INT4 (baseline currently unavailable)	INT4: registry.redhat.io/rhelai1/whisper-large-v3-turbo-quantized-w4a16:1.5
voxtral-mini-3b-2507	FP8 (baseline currently unavailable)	FP8: registry.redhat.io/rhelai1/voxtral-mini-3b-2507-fp8-dynamic:1.5
nvidia-nemotron-nano-9b-v2	FP8 (baseline currently unavailable)	FP8: registry.redhat.io/rhelai1/nvidia-nemotron-nano-9b-v2-fp8-dynamic:1.5

CHAPTER 5. VALIDATED RED HAT AI MODEL CAR CONTAINER IMAGES

You can use ModelCar container images to deploy validated models with Red Hat AI Inference. The following table lists the available ModelCar container images and their quantized variants.



NOTE
For minimum platform version requirements and validation status for each model, see [Model support matrix](#).

Table 5.1. Validated Red Hat AI ModelCar container images

Model	Quantized variants	ModelCar images
llama-4-scout-17b-16e-instruct	INT4, FP8	- Baseline: registry.redhat.io/rhelai1/modelcar-llama-4-scout-17b-16e-instruct:1.5 - INT4: registry.redhat.io/rhelai1/modelcar-llama-4-scout-17b-16e-instruct-quantized-w4a16:1.5 - FP8: registry.redhat.io/rhelai1/modelcar-llama-4-scout-17b-16e-instruct-fp8-dynamic:1.5
llama-4-maverick-17b-128e-instruct	FP8	- Baseline: registry.redhat.io/rhelai1/modelcar-llama-4-maverick-17b-128e-instruct:1.5 - FP8: registry.redhat.io/rhelai1/modelcar-llama-4-maverick-17b-128e-instruct-fp8:1.5

Model	Quantized variants	ModelCar images
mistral-small-3-1-24b-instruct-2503	INT4, INT8, FP8	● Baseline: registry.redhat.io/rhelai1/mod elcar-mistral-small-3-1-24b- instruct-2503:1.5 ● INT4: registry.redhat.io/rhelai1/mod elcar-mistral-small-3-1-24b- instruct-2503-quantized- w4a16:1.5 ● INT8: registry.redhat.io/rhelai1/mod elcar-mistral-small-3-1-24b- instruct-2503-quantized- w8a8:1.5 ● FP8: registry.redhat.io/rhelai1/mod elcar-mistral-small-3-1-24b- instruct-2503-fp8-dynamic:1.5
llama-3-3-70b-instruct	INT4, INT8, FP8	● Baseline: registry.redhat.io/rhelai1/mod elcar-llama-3-3-70b- instruct:1.5 ● INT4: registry.redhat.io/rhelai1/mod elcar-llama-3-3-70b-instruct- quantized-w4a16:1.5 ● INT8: registry.redhat.io/rhelai1/mod elcar-llama-3-3-70b-instruct- quantized-w8a8:1.5 ● FP8: registry.redhat.io/rhelai1/mod elcar-llama-3-3-70b-instruct- fp8-dynamic:1.5

Model	Quantized variants	ModelCar images
llama-3-1-8b-instruct	INT4, INT8, FP8	● Baseline: registry.redhat.io/rhelai1/modelcar-llama-3-1-8b-instruct:1.5 ● INT4: registry.redhat.io/rhelai1/modelcar-llama-3-1-8b-instruct-quantized-w4a16:1.5 ● INT8: registry.redhat.io/rhelai1/modelcar-llama-3-1-8b-instruct-quantized-w8a8:1.5 ● FP8: registry.redhat.io/rhelai1/modelcar-llama-3-1-8b-instruct-fp8-dynamic:1.5
granite-3-1-8b-instruct	INT4, INT8, FP8	● Baseline: registry.redhat.io/rhelai1/modelcar-granite-3-1-8b-instruct:1.5 ● INT4: registry.redhat.io/rhelai1/modelcar-granite-3-1-8b-instruct-quantized-w4a16:1.5 ● INT8: registry.redhat.io/rhelai1/modelcar-granite-3-1-8b-instruct-quantized-w8a8:1.5 ● FP8: registry.redhat.io/rhelai1/modelcar-granite-3-1-8b-instruct-fp8-dynamic:1.5

Model	Quantized variants	ModelCar images
phi-4	INT4, INT8, FP8	• Baseline: registry.redhat.io/rhelai1/modelcar-phi-4:1.5 • INT4: registry.redhat.io/rhelai1/modelcar-phi-4-quantized-w4a16:1.5 • INT8: registry.redhat.io/rhelai1/modelcar-phi-4-quantized-w8a8:1.5 • FP8: registry.redhat.io/rhelai1/modelcar-phi-4-fp8-dynamic:1.5
qwen2-5-7b-instruct	INT4, INT8, FP8	• Baseline: registry.redhat.io/rhelai1/modelcar-qwen2-5-7b-instruct:1.5 • INT4: registry.redhat.io/rhelai1/modelcar-qwen2-5-7b-instruct-quantized-w4a16:1.5 • INT8: registry.redhat.io/rhelai1/modelcar-qwen2-5-7b-instruct-quantized-w8a8:1.5 • FP8: registry.redhat.io/rhelai1/modelcar-qwen2-5-7b-instruct-fp8-dynamic:1.5

Model	Quantized variants	ModelCar images
mistral-small-24b-instruct-2501	INT4, INT8, FP8	- Baseline: registry.redhat.io/rhelai1/modelcar-mistral-small-24b-instruct-2501:1.5 - INT4: registry.redhat.io/rhelai1/modelcar-mistral-small-24b-instruct-2501-quantized-w4a16:1.5 - INT8: registry.redhat.io/rhelai1/modelcar-mistral-small-24b-instruct-2501-quantized-w8a8:1.5 - FP8: registry.redhat.io/rhelai1/modelcar-mistral-small-24b-instruct-2501-fp8-dynamic:1.5
mixtral-8x7b-instruct-v0-1	None	Baseline: registry.redhat.io/rhelai1/modelcar-mixtral-8x7b-instruct-v0-1:1.4
granite-3-1-8b-base	INT4 (baseline currently unavailable)	INT4: registry.redhat.io/rhelai1/modelcar-granite-3-1-8b-base-quantized-w4a16:1.5
granite-3-1-8b-starter-v2	None	Baseline: registry.redhat.io/rhelai1/modelcar-granite-3-1-8b-starter-v2:1.5
llama-3-1-nemotron-70b-instruct-hf	FP8	- Baseline: registry.redhat.io/rhelai1/modelcar-llama-3-1-nemotron-70b-instruct-hf:1.5 - FP8: registry.redhat.io/rhelai1/modelcar-llama-3-1-nemotron-70b-instruct-hf-fp8-dynamic:1.5

Model	Quantized variants	ModelCar images
gemma-2-9b-it	FP8	- Baseline: registry.redhat.io/rhelai1/mod elcar-gemma-2-9b-it:1.5 - FP8: registry.redhat.io/rhelai1/mod elcar-gemma-2-9b-it-fp8:1.5
deepseek-r1-0528	INT4 (baseline currently unavailable)	INT4: registry.redhat.io/rhelai1/mod elcar-deepseek-r1-0528- quantized-w4a16:1.5
qwen3-8b	FP8 (baseline currently unavailable)	FP8: registry.redhat.io/rhelai1/mod elcar-qwen3-8b-fp8- dynamic:1.5
kimi-k2-instruct	INT4 (baseline currently unavailable)	INT4: registry.redhat.io/rhelai1/mod elcar-kimi-k2-instruct- quantized-w4a16:1.5
gemma-3n-e4b-it	FP8	- Baseline: registry.redhat.io/rhelai1/mod elcar-gemma-3n-e4b-it:1.5 - FP8: registry.redhat.io/rhelai1/mod elcar-gemma-3n-e4b-it-fp8- dynamic:1.5
gpt-oss-120b	None	Baseline: registry.redhat.io/rhelai1/mod elcar-gpt-oss-120b:1.5
gpt-oss-20b	None	Baseline: registry.redhat.io/rhelai1/mod elcar-gpt-oss-20b:1.5

Model	Quantized variants	ModelCar images
qwen3-coder-480b-a35b-instruct	FP8 (baseline currently unavailable)	FP8: registry.redhat.io/rhelai1/modelcar-qwen3-coder-480b-a35b-instruct-fp8:1.5
whisper-large-v3-turbo	INT4 (baseline currently unavailable)	INT4: registry.redhat.io/rhelai1/modelcar-whisper-large-v3-turbo-quantized-w4a16:1.5
voxtral-mini-3b-2507	FP8 (baseline currently unavailable)	FP8: registry.redhat.io/rhelai1/modelcar-voxtral-mini-3b-2507-fp8-dynamic:1.5
nvidia-nemotron-nano-9b-v2	FP8 (baseline currently unavailable)	FP8: registry.redhat.io/rhelai1/modelcar-nvidia-nemotron-nano-9b-v2-fp8-dynamic:1.5
phi-4-reasoning	FP8 (baseline currently unavailable)	FP8: registry.redhat.io/rhai/modelcar-phi-4-reasoning-fp8-dynamic:3.0
qwen3-vl-235b-a22b-instruct-nvfp4	None	Baseline: registry.redhat.io/rhai/modelcar-qwen3-vl-235b-a22b-instruct-nvfp4:3.0
qwen3-next-80b-a3b-instruct	INT4 (baseline currently unavailable)	INT4: registry.redhat.io/rhai/modelcar-qwen3-next-80b-a3b-instruct-quantized-w4a16:3.0

Model	Quantized variants	ModelCar images
granite-4-0-h-tiny	FP8	- Baseline: registry.redhat.io/rhai/modelcar-granite-4-0-h-tiny:3.0 - FP8: registry.redhat.io/rhai/modelcar-granite-4-0-h-tiny-fp8-dynamic:3.0
granite-4-0-h-small	FP8	- Baseline: registry.redhat.io/rhai/modelcar-granite-4-0-h-small:3.0 - FP8: registry.redhat.io/rhai/modelcar-granite-4-0-h-small-fp8-dynamic:3.0
mistral-large-3-675b-instruct-2512	None	Baseline: registry.redhat.io/rhai/modelcar-mistral-large-3-675b-instruct-2512:3.0
mistral-large-3-675b-instruct-2512-nvfp4	None	Baseline: registry.redhat.io/rhai/modelcar-mistral-large-3-675b-instruct-2512-nvfp4:3.0
apertus-8b-instruct-2509	FP8 (baseline currently unavailable)	FP8: registry.redhat.io/rhai/modelcar-apertus-8b-instruct-2509-fp8-dynamic:3.0
nvidia-nemotron-3-nano-30b-a3b	FP8 (baseline currently unavailable)	FP8: registry.redhat.io/rhai/modelcar-nvidia-nemotron-3-nano-30b-a3b-fp8:3.0
ministral-3-14b-instruct-2512	None	Baseline: registry.redhat.io/rhai/modelcar-ministral-3-14b-instruct-2512:3.0

CHAPTER 6. VALIDATED MODELS FOR X86_64 CPU INFERENCE SERVING

The following large language models have been validated for use with Red Hat AI Inference on x86_64 CPUs.

Table 6.1. Validated models for inferencing with x86_64 CPU

Model	Hugging Face model card	Number of parameters
TinyLlama-1.1B-Chat-v1.0	TinyLlama/TinyLlama-1.1B-Chat-v1.0	1.1B
Llama-3.2-1B-Instruct	meta-llama/Llama-3.2-1B-Instruct	1B
granite-3.2-2b-instruct	ibm-granite/granite-3.2-2b-instruct	2B
TinyLlama-1.1B-Chat-v1.0-pruned2.4	RedHatAI/TinyLlama-1.1B-Chat-v1.0-pruned2.4	1.1B (pruned)



IMPORTANT

Quantization formats that require GPU-specific kernels, such as Marlin format, are not supported for CPU inference. Use AWQ or GPTQ quantization formats that are compatible with CPU execution.

The following table provides general guidance for approximate system RAM requirements based on model size:

Table 6.2. Memory requirements for inference serving with x86_64 CPU

Model size	Minimum RAM	Recommended RAM
125M - 500M	8 GB	16 GB
500M - 1B	16 GB	32 GB
1B - 3B	32 GB	64 GB



NOTE

Actual memory usage depends on the model architecture, context length, and batch size. Increase the **VLLM_CPU_KVCACHE_SPACE** environment variable to allocate more memory for the key-value cache when using longer context lengths.

CHAPTER 7. VALIDATED MODELS FOR USE WITH IBM POWER AND IBM SPYRE AI ACCELERATORS

The following large language models are supported for IBM Power systems with IBM Spyre AI accelerators.



NOTE

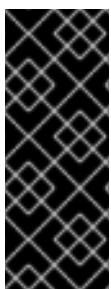
IBM Spyre AI accelerator cards support FP16 format model weights only. For compatible models, the Red Hat AI Inference inference engine automatically converts weights to FP16 at startup. No additional configuration is needed.

Table 7.1. IBM Granite models for use with IBM Spyre AI accelerators

Model	Hugging Face model card
granite-3.3-8b-instruct	ibm-granite/granite-3.3-8b-instruct
granite-embedding-30m-english	ibm-granite/granite-embedding-30m-english
granite-embedding-107m-multilingual	ibm-granite/granite-embedding-107m-multilingual
granite-embedding-125m-english	ibm-granite/granite-embedding-125m-english
granite-embedding-278m-multilingual	ibm-granite/granite-embedding-278m-multilingual

Table 7.2. Reranker models for use with IBM Spyre AI accelerators

Model	Hugging Face model card
bge-reranker-v2-m3	BAAI/bge-reranker-v2-m3



IMPORTANT

Pre-built IBM Granite models run with the specific Python packages that are included in the Red Hat AI Inference Spyre container image. The models are tied to fixed configurations for Spyre card count, batch size, and input/output context sizes.

Updating or replacing Python packages in the Red Hat AI Inference Spyre container image is not supported.

CHAPTER 8. VALIDATED MODELS FOR USE WITH IBM Z AND IBM SPYRE AI ACCELERATORS

The following large language models are supported for IBM Z systems with IBM Spyre AI accelerators.



NOTE

IBM Spyre AI accelerator cards support FP16 format model weights only. For compatible models, the Red Hat AI Inference inference engine automatically converts weights to FP16 at startup. No additional configuration is needed.

Table 8.1. Decoder models for use with IBM Spyre AI accelerators

Model	Hugging Face model card
granite-3.3-8b-instruct	ibm-granite/granite-3.3-8b-instruct
granite-3.3-8b-instruct-FP8	ibm-granite/granite-3.3-8b-instruct-FP8
granite-4.1-8b	ibm-granite/granite-4.1-8b
granite-4.1-8b-fp8	ibm-granite/granite-4.1-8b-fp8
Ministral-3-14B-Instruct-2512-BF16	mistralai/Ministral-3-14B-Instruct-2512-BF16



IMPORTANT

Pre-built IBM Granite models run with the specific Python packages that are included in the Red Hat AI Inference Spyre container image. The models are tied to fixed configurations for Spyre card count, batch size, and input/output context sizes.

Updating or replacing Python packages in the Red Hat AI Inference Spyre container image is not supported.

CHAPTER 9. VALIDATED MODELS FOR GEOSPATIAL INFERENCE WITH TERRATORCH

The following IBM and NASA Prithvi geospatial foundation models are validated for use with AI Inference and TerraTorch.



NOTE

Prithvi-EO-2.0 models use the Vision Transformer (ViT) architecture and require TerraTorch as the model implementation backend. These models accept GeoTIFF imagery as input and return segmentation predictions.

Table 9.1. Prithvi geospatial models for use with TerraTorch

Model	Use case	Hugging Face model card	Validated on
Prithvi-EO-2.0-300M-TL-Sen1Floods11	Flood detection and mapping	Prithvi-EO-2.0-300M-TL-Sen1Floods11	RHAIS 3.3
Prithvi-EO-2.0-300M-BurnScars	Burn scar detection	Prithvi-EO-2.0-300M-BurnScars	RHAIS 3.3

Explore the IBM and NASA geospatial models collection on [Hugging Face](#).



IMPORTANT

Prithvi geospatial models are validated for use with NVIDIA CUDA AI accelerators only.

These models require specific vLLM server arguments to function correctly. You must include **--skip-tokenizer-init**, **--enforce-eager**, and **--enable-mm-embeds** when starting the inference server.

For more information, see [Serving TerraTorch Models with vLLM](#).